

Advanced JavaScript Frameworks (Angular)
Student Management System

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Abstract:

The Student Management System is a web-based application developed to manage student records efficiently. It allows administrators to add, view, and edit student information such as name, email, course, and phone number. The system also provides a dashboard to display statistics like total students, active students, and courses. This application helps reduce manual work, improves data organization, and makes student information management faster and easier.

Objectives:

- To develop a Student Management System for managing student information efficiently.
- To provide a feature to add new student records into the system.
- To allow administrators to view the list of students in an organized format.
- To enable editing and updating student details when required.
- To display student statistics through a dashboard for better monitoring.
- To reduce manual paperwork and improve data management.

Scope of the Project:

- The Student Management System is designed to simplify the process of managing student information in an educational institution. The system allows administrators to store, view, add, and update student details in an organized manner. It provides a centralized platform where all student records can be maintained efficiently.
- The project helps reduce manual work and improves accuracy in managing student data. It also provides a dashboard to monitor student statistics such as total students, active students, and courses. The system can be further expanded in the future by adding features like student login, attendance management, and result management.

Tools & Technologies Used:

Tool / Technology	Purpose
Angular (v20)	Frontend framework used to build the web application
TypeScript	Used for writing the application logic and improving code
Angular Material	Provides UI components and styling for a better user interface.
JSON Server	Used as a mock backend to store and retrieve student data.
RxJS	Handles asynchronous operations and data streams in Angular.
HTML5	Used to create the structure of web pages.
CSS3	Used for styling and layout of the application.
Node.js	Runtime environment required to run Angular applications.
Angular CLI	Tool used for creating, building, and managing Angular projects
VS Code	Code editor
Chrome DevTools	Testing & debugging

Application Overview

- The Student Management System is a web-based application developed to manage student information efficiently. It allows administrators to add, view, and edit student details such as name, email, course, and phone number. The system also provides a dashboard that displays student statistics like total students, active students, available courses, and graduated students.
- The application is built using Angular, TypeScript, HTML, and CSS, with Angular Material for the user interface and JSON Server as a mock backend. It provides a simple and user-friendly interface that helps institutions manage student records digitally and reduce manual work.

Module Description:

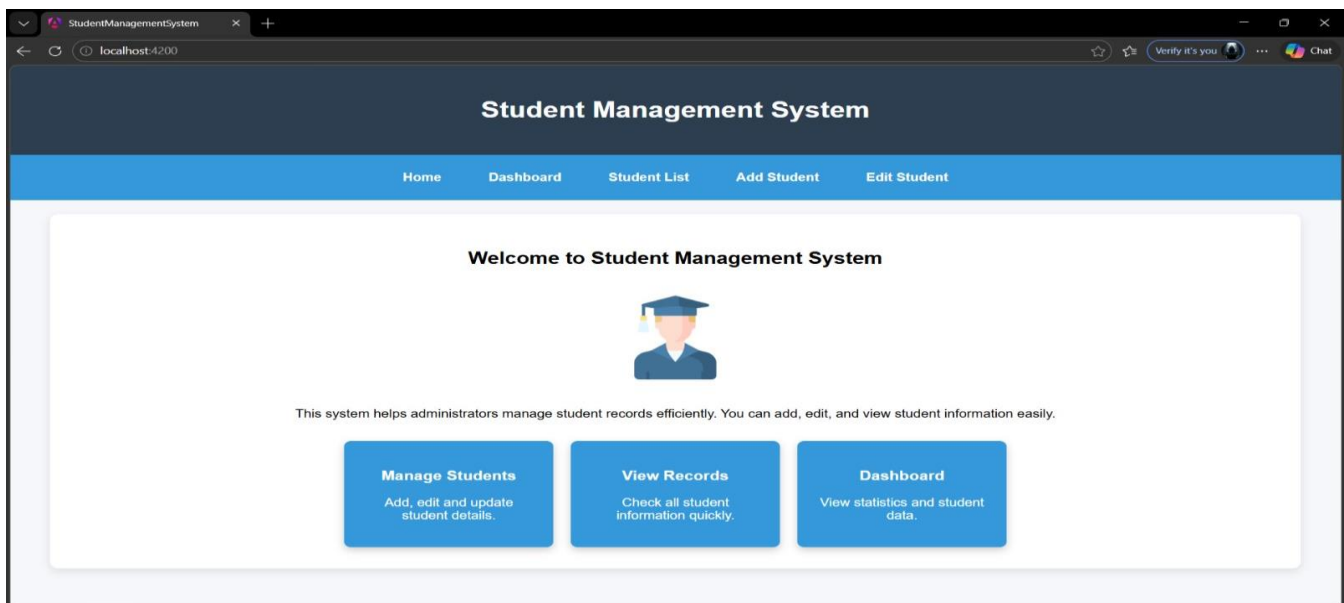
1 Dashboard Module

- The Dashboard Module serves as the central overview component of the Student Management System. It provides a summarized view of important student-related information, allowing administrators to quickly understand the current status of student records in the system.
- This module displays key information such as the total number of students, active students, available courses, and graduated students. The data is retrieved from the backend using Angular services and RxJS observables, ensuring that the displayed statistics remain accurate and updated.
- The Dashboard Module improves usability by presenting important student statistics in a clear and organized format. By combining all essential information into a single interface, it helps administrators monitor the system easily and manage student records more efficiently.

Key Functions:

- Displays a summary of student statistics such as total students, active students, available courses, and graduated students.
- Provides a quick overview of the student management system.
- Retrieves and displays updated student data from the backend.
- Helps administrators monitor student records efficiently.
- Provides easy navigation to other modules like Student List and Add Student.

Screenshot :





Student List Module:

- The Student List Module is responsible for displaying and managing all student records within the system. It allows administrators to view detailed student information including student name, email, course, and phone number.
- This module interacts with the backend service to retrieve student data and display it in a structured table format. Angular components and data binding are used to dynamically render student records and ensure that any updates are reflected immediately.
- The Student List Module plays an important role in maintaining organized student data, enabling administrators to easily view and manage all students.

Key Functions:

- Displays a list of all students
- Shows student details such as name, email, course, and phone number
- Allows administrators to view student records
- Provides edit options for modifying student information
- Ensures accurate and updated student data

Student List

ID	Name	Email	Course	Phone
1	Virat Kohli	virat@college.edu	BCA	9876543210
2	Rahul Sharma	rahul@college.edu	MCA	9123456780
3	Anjali Mehta	anjali@college.edu	BSc CS	9988776655
4	Rohit Verma	rohit@college.edu	BBA	9876501234
5	Sneha Reddy	sneha@college.edu	BCA	9876604321
6	Karan Patel	karan@college.edu	MCA	9123498765
7	Priya Nair	priya@college.edu	BSc IT	9090909090
8	Arjun Singh	arjun@college.edu	BCA	9898989898
9	Meera Joshi	meera@college.edu	BBA	9111111111

Add Student Module:

- The Add Student Module allows administrators to add new student records to the system. This module provides a form where users can enter student information such as name, email, course, and phone number.
- The form uses Angular form validation to ensure that the entered information is correct and complete before it is submitted. Once the data is submitted, it is stored in the backend database using Angular services.
- This module helps institutions maintain an updated and organized student database by allowing quick and easy addition of new students.

Key Functions:

- Provides a form to add new student details
- Collects information such as name, email, course, and phone number
- Validates student data before submission
- Stores student information in the backend
- Ensures proper and structured student record creation

Student Management System

[Home](#)[Dashboard](#)[Student List](#)[Add Student](#)[Edit Student](#)

Add Student

Name**Email****Course****Phone**



Edit Student Module:

- The Edit Student Module allows administrators to modify existing student information. When the administrator selects a student record from the student list, the system loads the student details into an editable form.
- The administrator can update details such as student name, email, course, or phone number, and the changes are saved back to the database.
- This module ensures that student information remains accurate and up to date.

Key Functions:

- Allows updating existing student records
- Retrieves selected student data for editing
- Updates information such as name, email, course, and phone number
- Saves updated data to the backend
- Maintains accurate student records

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Student Management System

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Edit Student

ID	Name	Email	Course	Phone	Action
1	Virat Kohli	virat@college.edu	BCA	9876543210	Edit
2	Rahul Sharma	rahul@college.edu	MCA	9123456780	Edit
3	Anjali Mehta	anjali@college.edu	BSc CS	9988776655	Edit
4	Rohit Verma	rohit@college.edu	BBA	9876501234	Edit
5	Sneha Reddy	sneha@college.edu	BCA	9876604321	Edit
6	Karan Patel	karan@college.edu	MCA	9123498765	Edit
7	Priya Nair	priya@college.edu	BSc IT	9090909090	Edit
8	Arjun Singh	arjun@college.edu	BCA	9898989898	Edit

UI & Styling Strategy

- Angular Material components for consistent UI design.
- MatTable for displaying employee and attendance records.
- MatFormField, MatInput, and MatSelect for forms.
- MatSnackBar for notifications and alerts.
- Responsive layout using Material toolbar and side navigation.

Angular & TypeScript Overview:

- TypeScript interfaces define structured models for Student data.
- Services handle CRUD operations using HttpClient.
- RxJS Observables manage asynchronous data flow.
- Routing modules enable navigation between different pages.
- Data binding and directives ensure dynamic UI updates.

Key Features:

- Centralized student information management
- Dashboard showing student statistics
- Ability to add, view, and edit student records
- Organized display of student data in tables
- Responsive and user-friendly interface using Angular Material
- Efficient management of student information

Challenges Faced & Solutions:

- **Environment setup issues:** Resolved by configuring PowerShell execution policies and properly installing Angular CLI.
- **Understanding project structure:** Solved by organizing code into models, services, and components.
- **Component generation errors:** Fixed by running Angular CLI commands inside the correct project directory.
- **Backend integration difficulties:** Addressed using JSON Server to simulate REST APIs.
- **UI consistency challenges:** Overcome by using Angular Material for a uniform and responsive interface.

Outcome:

- Successfully developed a functional student management system using Angular
- Implemented dashboard, student list, add student, and edit student modules
- Achieved a responsive and user-friendly interface



Future Enhancements:

- Add user authentication and login system
- Add student result and grade management
- Connect to a real backend database
- Generate reports and analytics for students



Conclusion:

The Student Management System was successfully designed and implemented using Angular and TypeScript. The application demonstrates the use of component-based architecture, services, routing, and form handling to manage student records effectively. By integrating Angular Material and JSON Server, the system provides a responsive and user-friendly interface. This project improves understanding of modern web development and provides a strong foundation for building scalable educational management systems.



References:

- Angular Documentation – <https://angular.dev>
- Angular Material Documentation – <https://material.angular.io>
- TypeScript Documentation – <https://www.typescriptlang.org>
- JSON Server Documentation – <https://github.com/typicode/json-server>
- Node.js Official Website – <https://nodejs.org>

